NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Pathology: Restrictive Lung Diseases – Idiopathic Pulmonary Fibrosis and other Pulmonary Fibrosing Diseases

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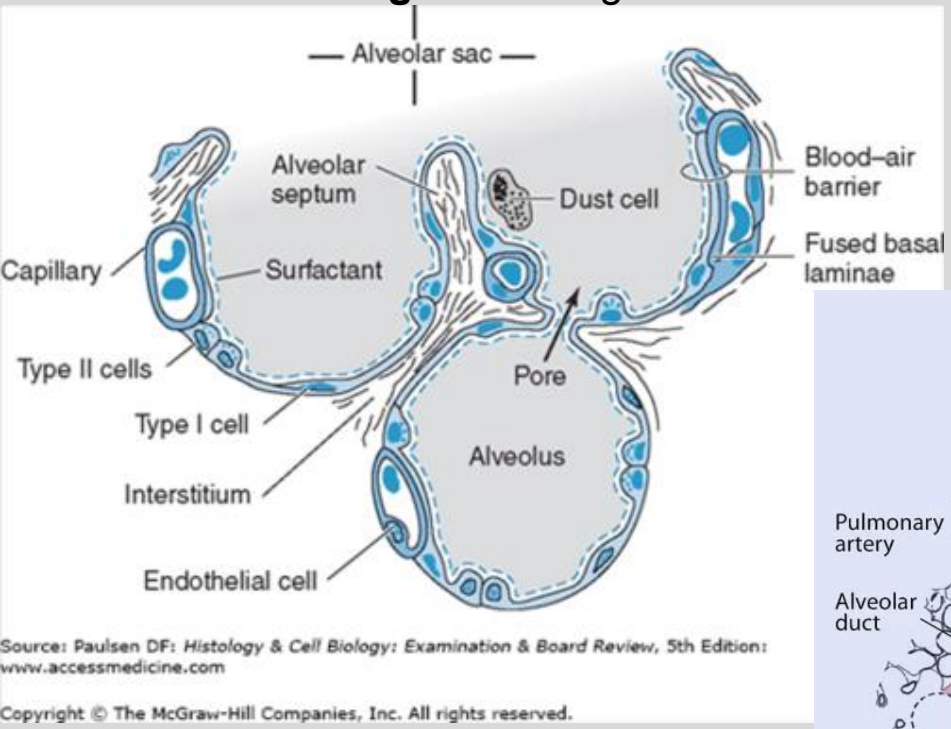
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Session Objectives

1. Examine the general pathophysiological and clinicopathological features of **chronic diffuse interstitial lung disease** and apply this to explain the disease of idiopathic pulmonary fibrosis (**IPF**)
2. Interpret the histological pattern of usual interstitial pneumonia (**UIP**)as it relates to IPF and explain its correlation with gross findings.
3. Appraise proposed mechanisms of pulmonary fibrosis including the contribution of **telomerase mutations**.
4. Compare non-specific interstitial pneumonia (**NSIP**) with IPF.
5. Distinguish **smoking** related interstitial diseases which includes desquamative interstitial pneumonia (**DIP**) and respiratory bronchiolitis-associated interstitial lung disease (**RBILD**).

Interstitial (Infiltrative, Restrictive) Lung Disease (ILD)

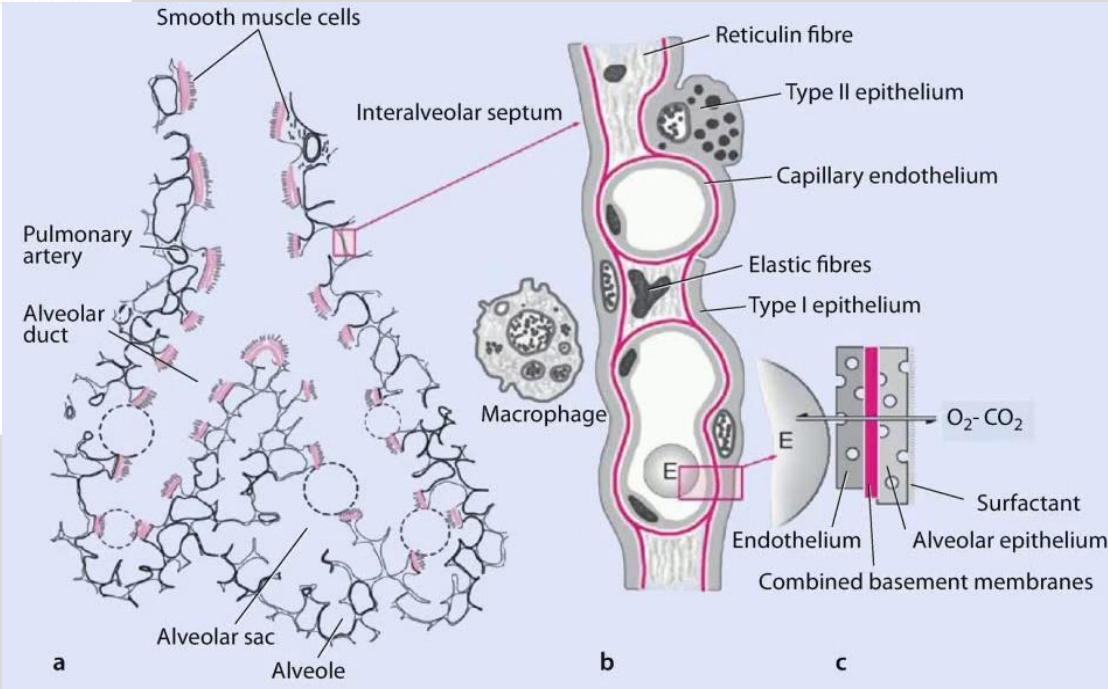
- **Any** process that results in **inflammatory-fibrotic infiltration** of the **alveolar septa/interstitium** producing effects on the capillary endothelium & alveolar epithelium & includes any condition that leads to **abnormal healing** response leading to tissue **fibrosis/scarring** in the lungs can be classified as interstitial lung disease



Source: Paulsen DF: *Histology & Cell Biology: Examination & Board Review*, 5th Edition: www.accessmedicine.com

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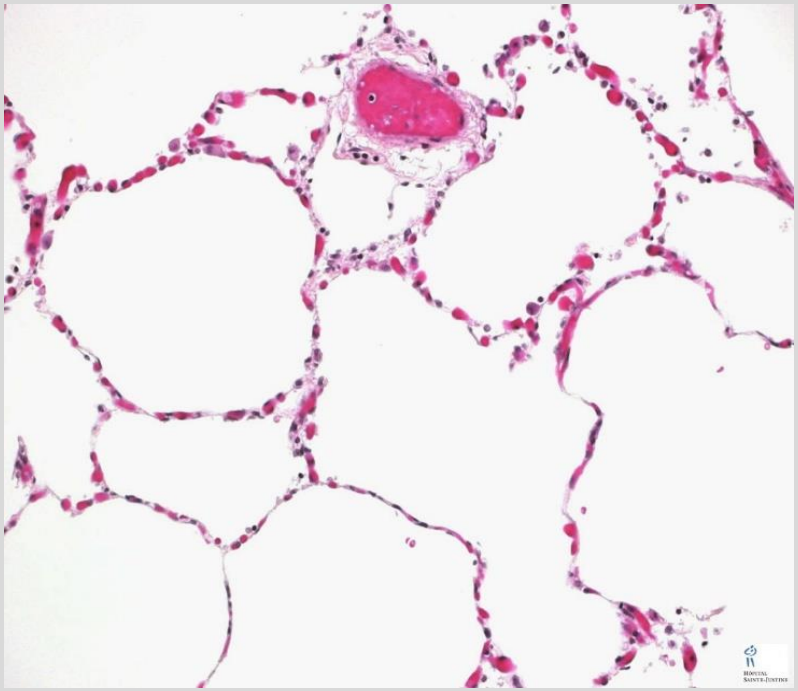


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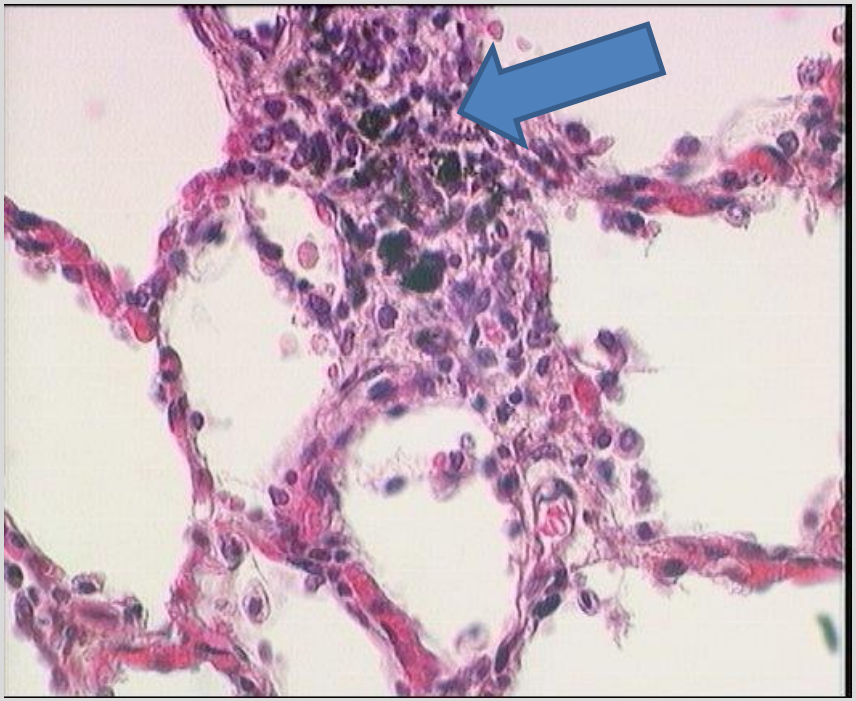
H&E stains: ILD

Diffuse or patchy, usually **chronic** (lymphocytes & macrophages, fibrosis), involvement of pulmonary connective tissue, mainly peripheral lungs & delicate interstitium in the alveolar walls

Normal lung alveoli



Interstitial Lung Disease

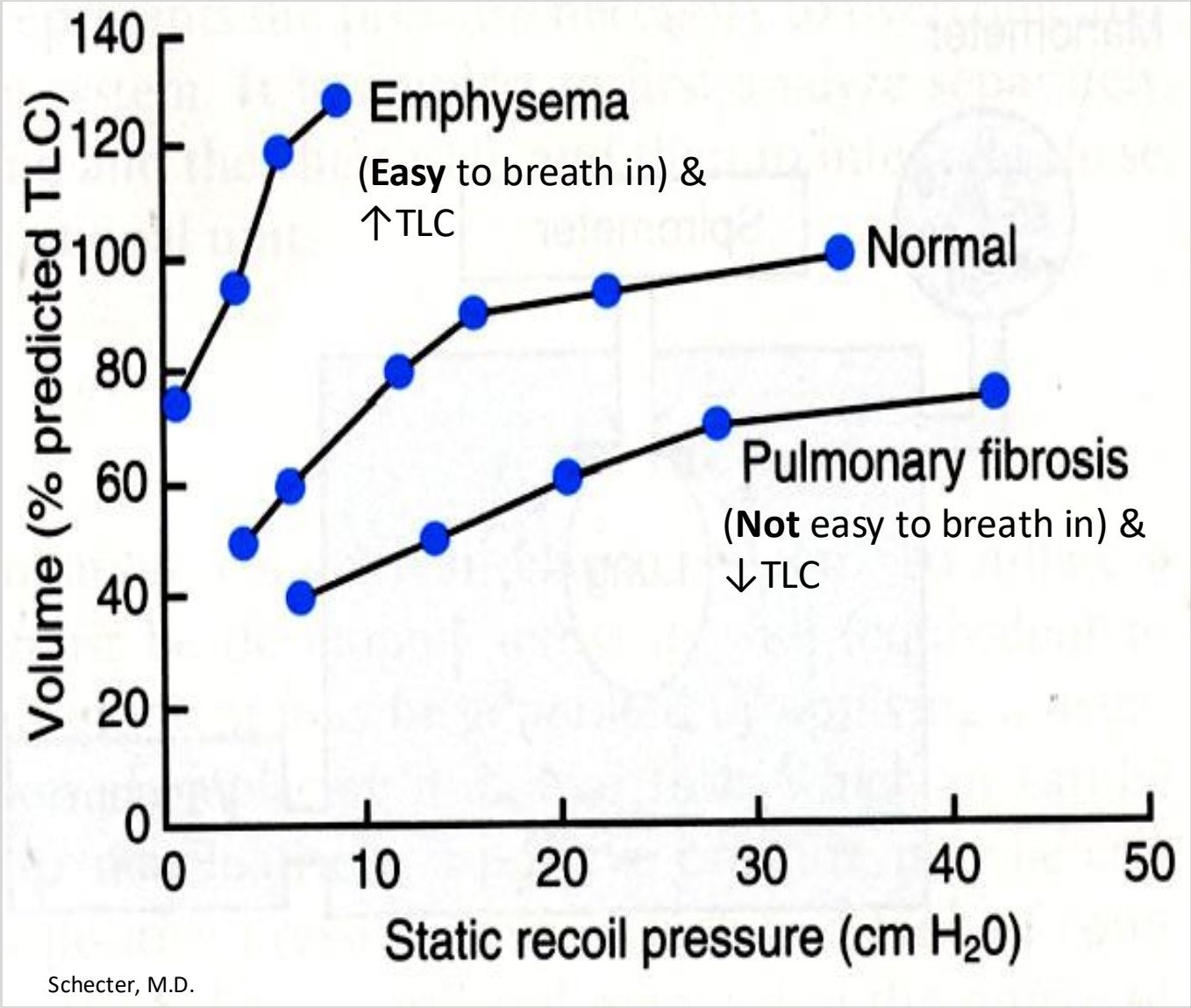


https://www.humpath.com/spip.php?article3950

Chronic ILD

- About 15% of non-infectious diseases seen by pulmonary physicians
- Signs/Symptoms (**Sx/Sx**): Dyspnea, tachypnea, end-inspiratory crackles
- **Lab** findings:
  - ↓ in diffusion capacity (DLCO), decrease in forced vital capacity (FVC)
  - ↓ in lung volume
  - ↓ in lung compliance
  - Chest x-ray may show ground glass (hazy gray area on x-ray) shadows
- End-stage : "**Honey-comb**" lung

- Fibrosis (restrictive) **can't** inhale vs. COPD (obstructive) **can't** exhale



INTERSTIAL LUNG DISEASES – only note the diversity, this focus is on idiopathic types

Connective Tissue Diseases
Scleroderma
Polymyositis-Dermatomyositis
Systemic Lupus Erythematosus
Rheumatoid Arthritis
Mixed Connective Tissue Disease
Ankylosing Spondyitis

Primary (Unclassified) - another lecture
Sarcoidosis
Langerhans cell histiocytosis
Amyloidosis
Pulmonary vasculitis
Lipoid pneumonia
Lymphangitic carcinomatosis
Bronchoalveolar carcinoma
Pulmonary lymphoma
Gaucher's Disease
Niemann-Pick Disease
Hermansky-Pudlak syndrome
Neurofibromatosis
Lymphangioleiomyomatosis
Tuberous Sclerosis
ARDS
AIDS
Bone Marrow Transplantation
Postinfectious
Eosinophilic pneumonia
Alveolar Proteinosis
Diffuse Alveolar Hemorrhage Syndromes
Alveolar microlithiasis
Metastatic calcification

Treatment-Related / Drug-Induced
Antibiotics – nitrofurantoin, sulfasalazine
Antiarrhythmics – amiodarone, propranolol
Anti-inflammatories – gold, penacillamine
Anticonvulsants – dilantin
Chemotherapeutic agents – bleomycin, cyclophosphamide, methotrexate, azathioprine
Therapeutic radiation
Oxygen toxicity
Narcotics

Occupational and Environmental Diseases - another lecture

Inorganic

Silicosis
Asbestosis
Hard-metal pneumoconiosis
Coal worker's pneumoconiosis
Berylliosis
Aluminum oxide fibrosis
Talc pneumoconiosis
Siderosis (arc welder)
Stannosis (tin)

Organic

Bird breeder's lung
Farmer's lung
Bacteria – e.g. NTB mycobacteria
Fungi – e.g. Aspergillus
Animal protein – e.g. Avian
Chemical sensitizers -
e.g. isocyanates

Idiopathic Fibrotic Disorders
Acute interstitial pneumonitis (Hamman-Rich syndrome)
Idiopathic Pulmonary Fibrosis
Familial Idiopathic Pulmonary Fibrosis
Desquamative interstitial pneumonitis
Respiratory bronchiolitis
Cryptogenic organizing pneumonia
Nonspecific interstitial pneumonitis
Lymphocytic interstitial pneumonia (Sjögrens Syndrome, AIDS, Hashimoto's)
Autoimmune pulmonary fibrosis (inflammatory bowel disease, PBC, ITP, AIHA)

ATS/ERS (European Respiratory Society (ERS) & American Thoracic Society (ATS) Task Force)

Definition of **Idiopathic Pulmonary Fibrosis** & some clinical features

- **Unknown** etiology **limited** to the lungs
  - Distinct chronic fibrosing interstitial pneumonia
  - Immunologic, viral, and genetic factors may play role
- A type of **chronic** fibrosing **interstitial** pneumonia & a **clinicopathological syndrome** w/ progressive IPF & respiratory failure, & histopathological pattern of **UIP**

- Most **common** & **lethal** form of pulmonary fibrosis: Median survival 3.8 years after diagnosis – lung transplant only definitive therapy

- More common in **elderly men**, with most cases presenting over 60 years of age, particularly when there's familial risk for pulmonary fibrosis

- Presentation Sx/Sx: **Insidious progressive** onset of otherwise **unexplained** dyspnea on exertion, duration of illness >3 months, bibasilar inspiratory crackles

ATS/ERS. Am J Respir Crit Care Med. 2002;165:277-304.

Typical Clinical Presentation

- Middle aged male, 50-70 years of age
- Occupational & exposure history
- Most have Sx/Sx for 12-18 months prior to definitive evaluation

- **New onset** of progressive exertional dyspnea and non-productive **cough**
- Constitutional symptoms are uncommon, but may see weight loss, fever, fatigue occasionally present (rash, arthritis, myositis should suggest an alternate diagnosis)

- Physical exam: **Bibasilar** late inspiratory fine crackles (Velcro rales), tachypnea, cyanosis, clubbing in 40-75% late in disease course, cardiac exam usually normal until middle-late stages (augmented P2, right-sided heave, S3 gallop)

Diagnosing IPF

- Diagnosis (Dx) by:
  - Corresponds to the **histopathologic** or **imaging** pattern of usual interstitial pneumonia (**UIP**) &
  - **Exclusion/Absence** of alternative causes such as drug toxicity, environmental exposure (ex asbestos) or collagen vascular disease (ex. scleroderma, rheumatoid arthritis)

- Abnormal results of pulmonary function tests (**PFT**): evidence of restriction (ex. ↓ vital capacity (VC), often normal or slightly↑ **FEV1/FVC ratio**) & impaired gas exchange (ex. ↓ DLCO)



CT image of chest

- **CT** imaging findings complement the histology
  - UIP-pattern of fibrosis is characterized by **honeycombing** cysts and reticular septal thickening with **subpleural** and **posterior basal** predominance (bibasilar reticular abnormalities with minimal ground-glass opacities)

ATS/ERS Consensus Statement. *Am J Respir Crit Care Med*. 2002;165:277-304.

~Etiologies & pathogenesis of IPF

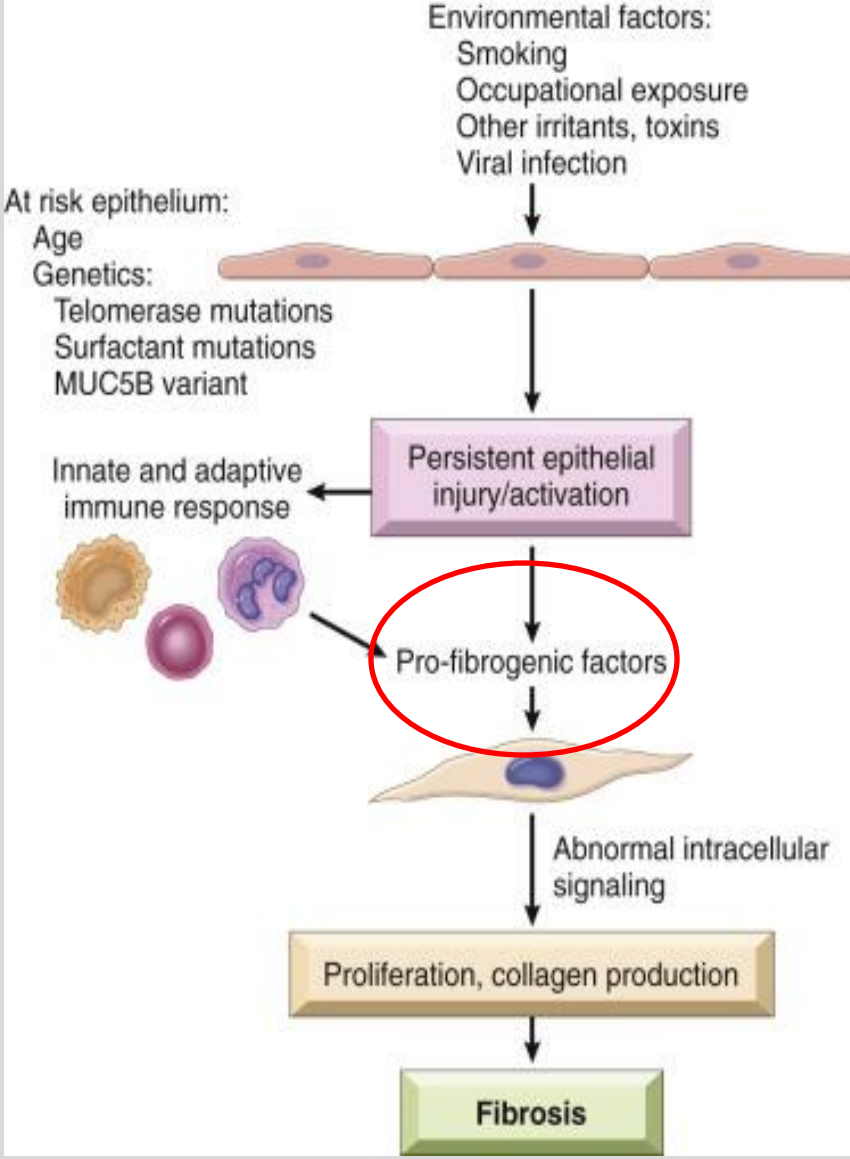
- Exact cause is unknown

- **Genetic** role suggested by familial IPF
  - Up to 20% of patients with IPF have a **family history** of interstitial lung disease (ILD) or pulmonary fibrosis, & fibroblasts seem to demonstrate neoplastic or neoproliferative characteristics
  - Fibrosis arises in genetically predisposed people prone to **aberrant repair** of recurrent alveolar epithelial cell injuries by environmental **exposures** (e.g., smoking, metal fumes, wood dust, occupational exposure, etc.)
  - Injured epithelial cells may be a source of **profibrogenic factors**, such as **TGF-beta** and innate & adaptive immune cells produce profibrogenic factors as host response to epithelial cell damage
    - Association with concurrent or previous history of **smoking** in 60% of patients
    - History of flulike illness favors **viral** origin in some patients

- 20% of cases are associated w/ **collagen vascular disease** including RA, SLE and progressive systemic sclerosis – suggests **immune** system involvement

• IPF etiology ~ persistent epithelial injury

- Factors secreted from injured/activated epithelium activate interstitial fibroblasts
  - These **activated fibroblasts** show signaling abnormalities that lead to increased signaling through the PI3K/AKT pathway
- Activated fibroblasts synthesize and deposit **collagen**, leading to interstitial fibrosis and eventual respiratory failure



Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Ch.15, fig. 15.13

Familial IPF

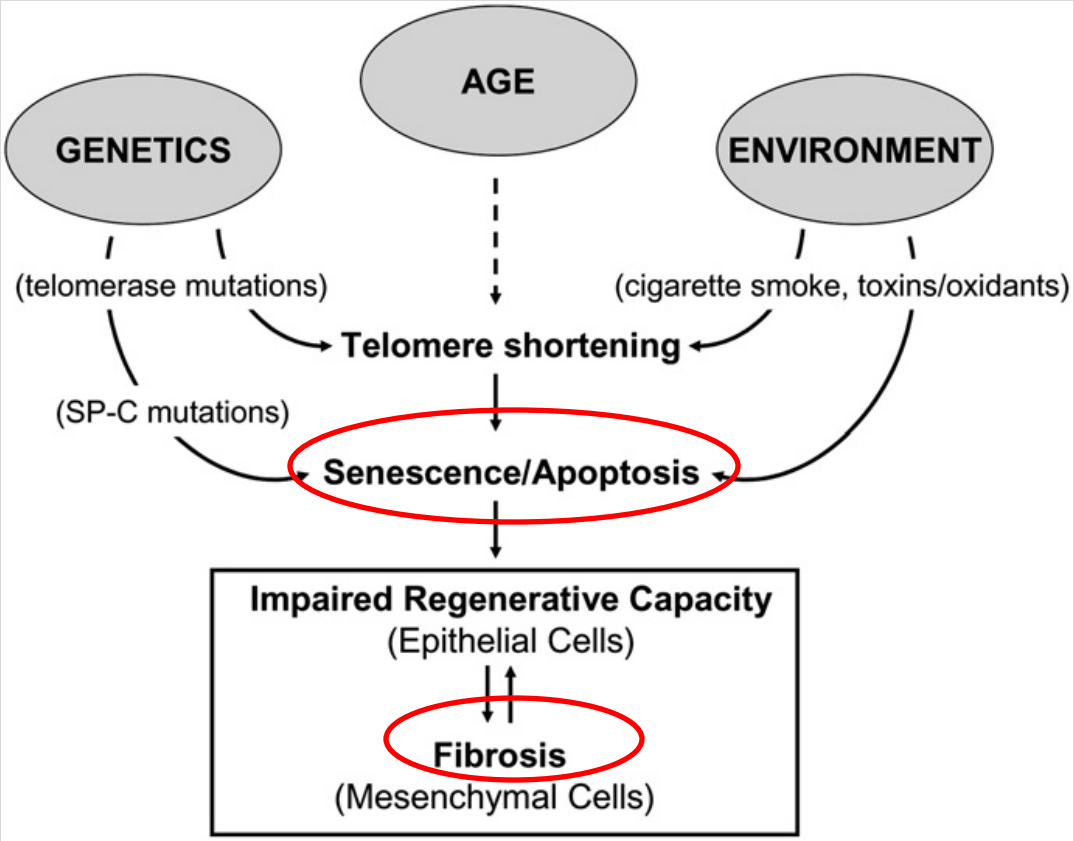
- **~3%** of cases cluster in families
  - Those with IPF had **mutant telomerase** & short telomeres
  - Demonstrated that mutation has **autosomal dominant** inheritance w/ variable penetrance

  - Proposed that **fibrosis** occurs due to **death of alveolar cells**
  - Associated with ↑ risk for IPF
    - Family studies found mutations in **TERT & TERC** genes encode parts of telomerase enzyme

- **Other genes** involved with maintenance of telomeres: Mutations in PARN, and RTEL1 genes are and the following:
  - About 1/3 of IPF cases are associated with single-nucleotide polymorphism in promoter MUC5B (member of mucin family) which increases secretion of MUC5B and may alter **mucociliary clearance**
  - 15% of familial IPF associated with inherited defects in genes **maintaining telomeres**
  - 25% of sporadic IPF associated with abnormal telomere shortening in peripheral blood **lymphocytes**
  - Other rare familial forms of IPF are associated with mutations in genes encoding parts of **surfactant**

Armanios et al, NEJM 2007
Robbins and Cotran, Pathologic Basis of Disease, 11th edition, 2026, Ch. 15

IPF proposed etiology



- Patients **w/ IPF** can have **normal** telomerase but shortened telomeres
- & individuals w/ shortened telomeres can be **asymptomatic**

- Multiple elements impact telomere length & function - ex. surfactant protein C (**SP-C**) is one of the pulmonary surfactant proteins (in humans this is encoded by the SFTPC gene)

Thannickal VJ, Lloyd JE. Am J Respir Crit Care Med. 2008;178:663-665.

Pathologic evaluation for IPF: Surgical lung biopsy

- If clinical & radiographic findings are atypical for IPF
  - **UIP**/IPF **pattern**
    - Fibroblastic foci
    - Temporal heterogeneity
    - Honeycombing appearance
    - Minimal inflammation
- Transbronchial biopsy not useful for diagnosis, but can be used to **exclude** other diagnoses

Adapted from ATS/ERS. Am J Respir Crit Care Med. 2000;161:646-664

IPF pathology

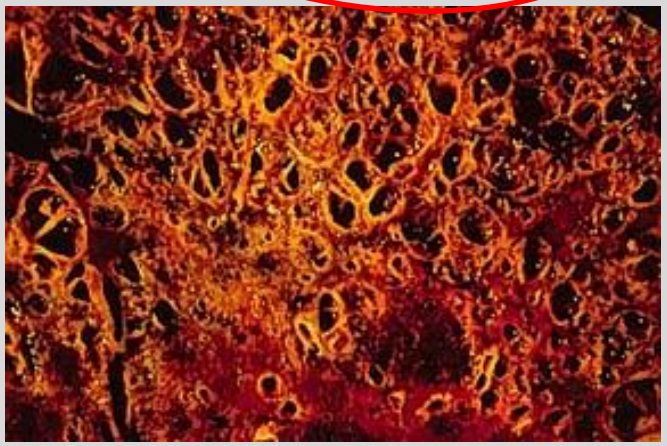
Lungs cut surface: **Fibrosis**, especially (esp.) in lower lobes w/ distinctive distribution in **subpleural** regions along the **interlobular septa & honeycombing**





Cobblestoned pleural surfaces (nodular) resulting from retraction of scars (fibrosis) along interlobular septa



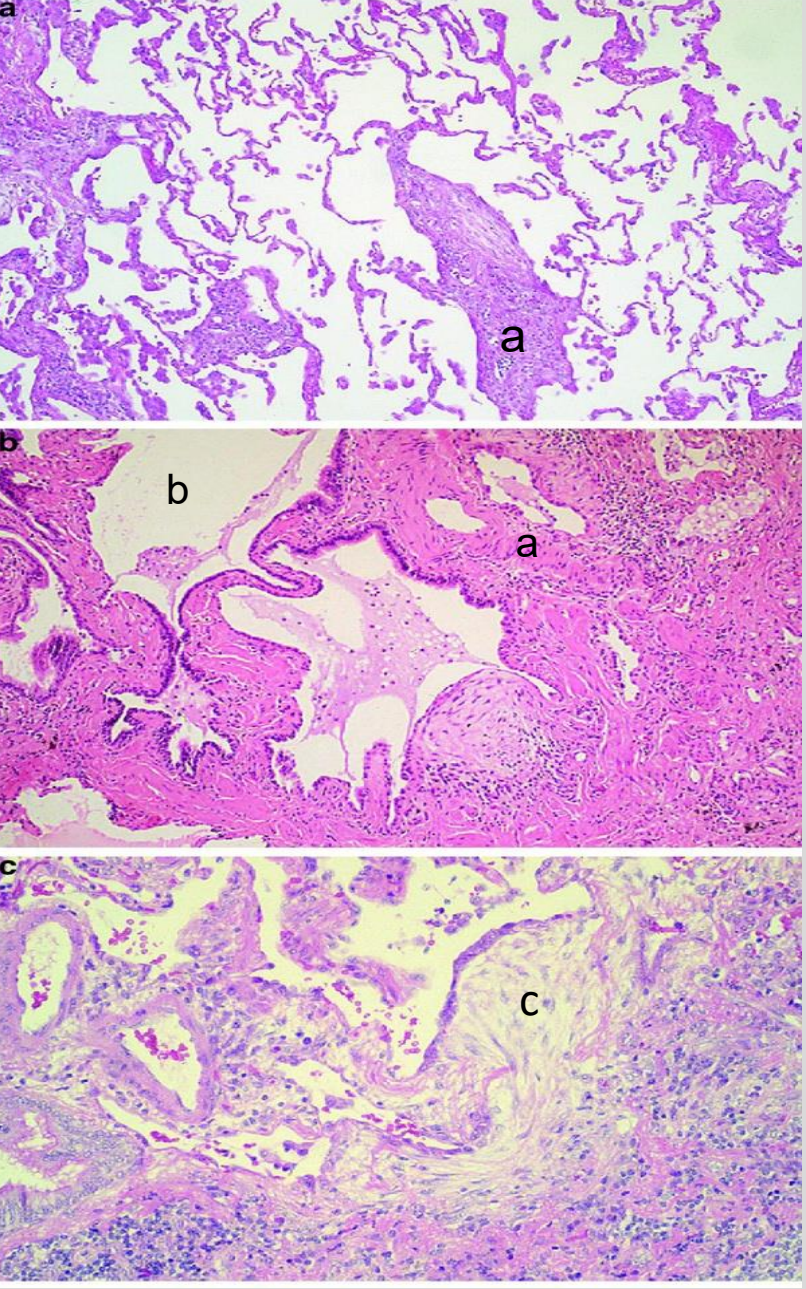


http://medlibes.com/entry/idiopathic-pulmonary-fibrosis
http://www.slideshare.net/ActionforPF/ipf-webinar-for-the-british-lung-foundation-2012-dr-helen-parfrey
https://lh3.googleusercontent.com/CEj8KZ0FGghITNomSh_V6CGAbN0RvPT9fz_OryQhlly86LYGzu5mnH6dVZEjWrNO_Sf5oQ=s85
Robbins and Cotran, Pathologic Basis of Disease, 11th edition, 2026, Ch. 15

IPF histopathology

Pattern of fibrosis is "**usual interstitial pneumonia (UIP)**"

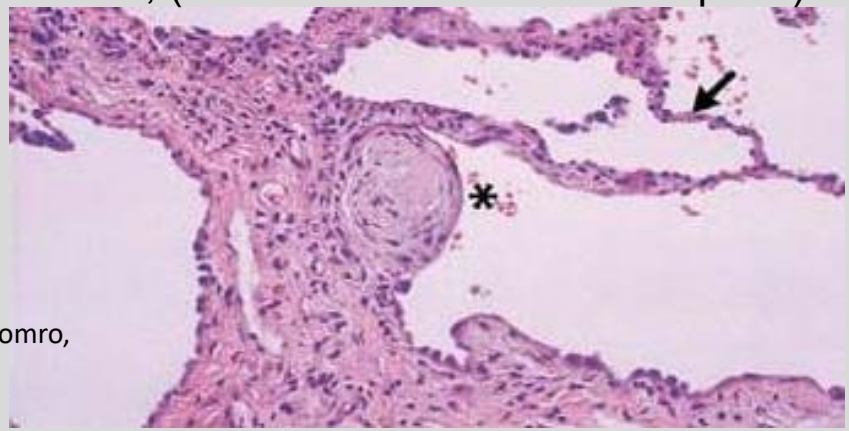
- **Patchy interstitial fibrosis ("spatial heterogeneity")**, varies in intensity & age ("**temporal heterogeneity**")
  - Early & late lesions **co-exist**
    - **Early** lesions show **fibroblastic foci**
      - Mild to moderate inflammation, mostly lymphocytes, few plasma cells, few neutrophils
    - **Later**: Dense fibrosis results in alveolar wall collapse & **cystic space** formation lined by hyperplastic type II pneumocytes or bronchiolar epithelium (**honeycomb fibrosis**)



UIP pattern (H&E Stains): Low-power – areas of dense fibrosis, with collapse & obliteration of alveolar air spaces

- **Spatial heterogeneity**, exemplified by dense scarring adjacent to relatively spared alveoli
- No necrosis or substantial inflammation
- Temporal heterogeneity
  - (a) **patchy fibrosis**[a] with intervening normal lung parenchyma
  - (b, c) marked fibrosis with collagenous scarring, **honeycomb**[b] changes and **fibroblastic foci**[c] **(temporal heterogeneity)**

High-power: **Fibroblast focus*** is visible as a nodule of spindle cells against a pale-staining extracellular matrix, (**arrow** = normal alveolar septum)



Gross T and Hunninghake G. N Engl J Med 2001;345:517-525

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Rajni Chibbar, Francis Shih, Monica Baga, Emina Torlakovic, Kumar Ramlall, Robert Skomro, Donald W Cockcroft and Edmond G Lemire
Modern Pathology (2004) 17, 973–980

Why do patients die from IPF?

- Acute exacerbations
- Cor pulmonale
- Pulmonary infection
- Cardiovascular Disease
- Pulmonary Hypertension
- Lung Cancer

Non-Specific Interstitial Pneumonia (NSIP)

- Idiopathic or associated with connective tissue disease
- **No fibroblastic foci, no honeycombing, no hyaline membranes, no granulomas**
- Better prognosis

- Tend to be **female**, **non-smokers**
- Sx/Sx Dyspnea, cough
- CT – **Bilateral** symmetrical lower lobe reticular fibrosis, ”ground glass” opacities

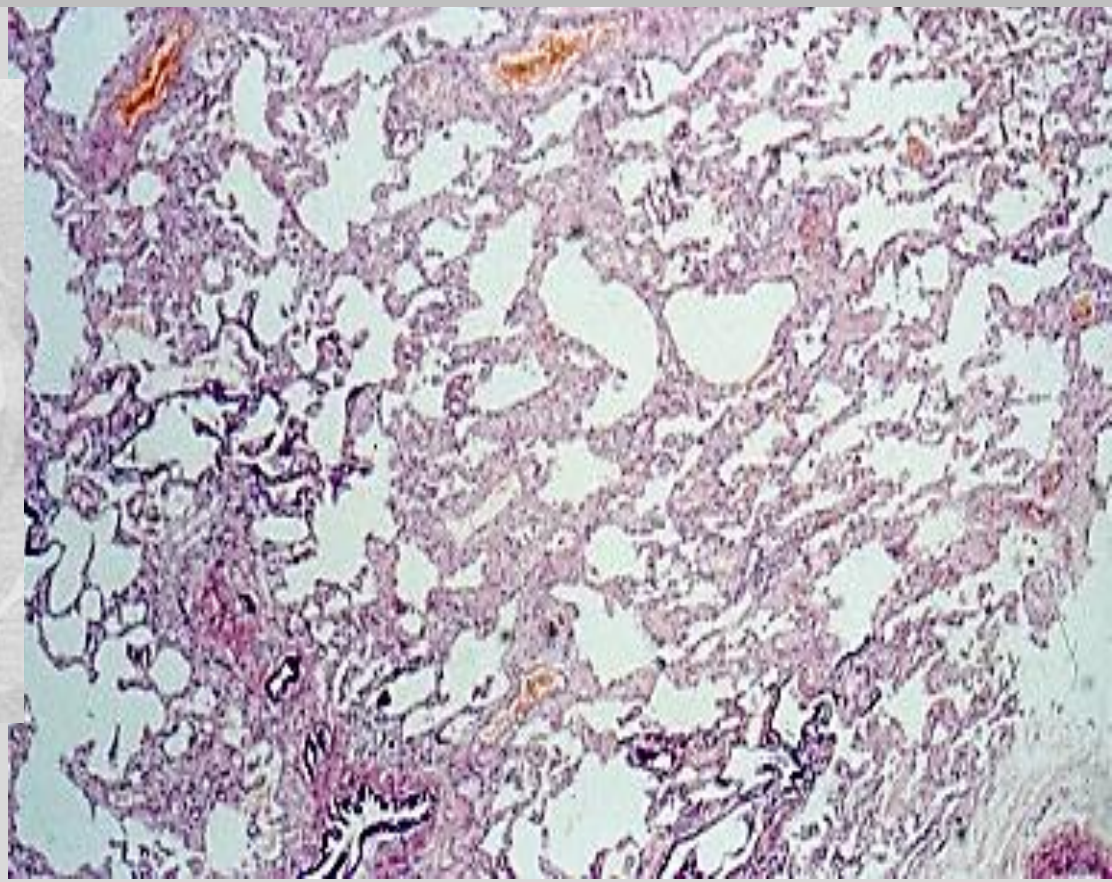
- **Cellular pattern** – mild – moderate chronic interstitial inflammation (lymphocytes, some plasma cells) in uniform or patchy distribution
- **Fibrosing pattern** – Diffuse or patchy interstitial fibrotic lesions at roughly same stage of development (important **distinction from UIP**)

“Classical” Variant of NSIP



Chest CT w/

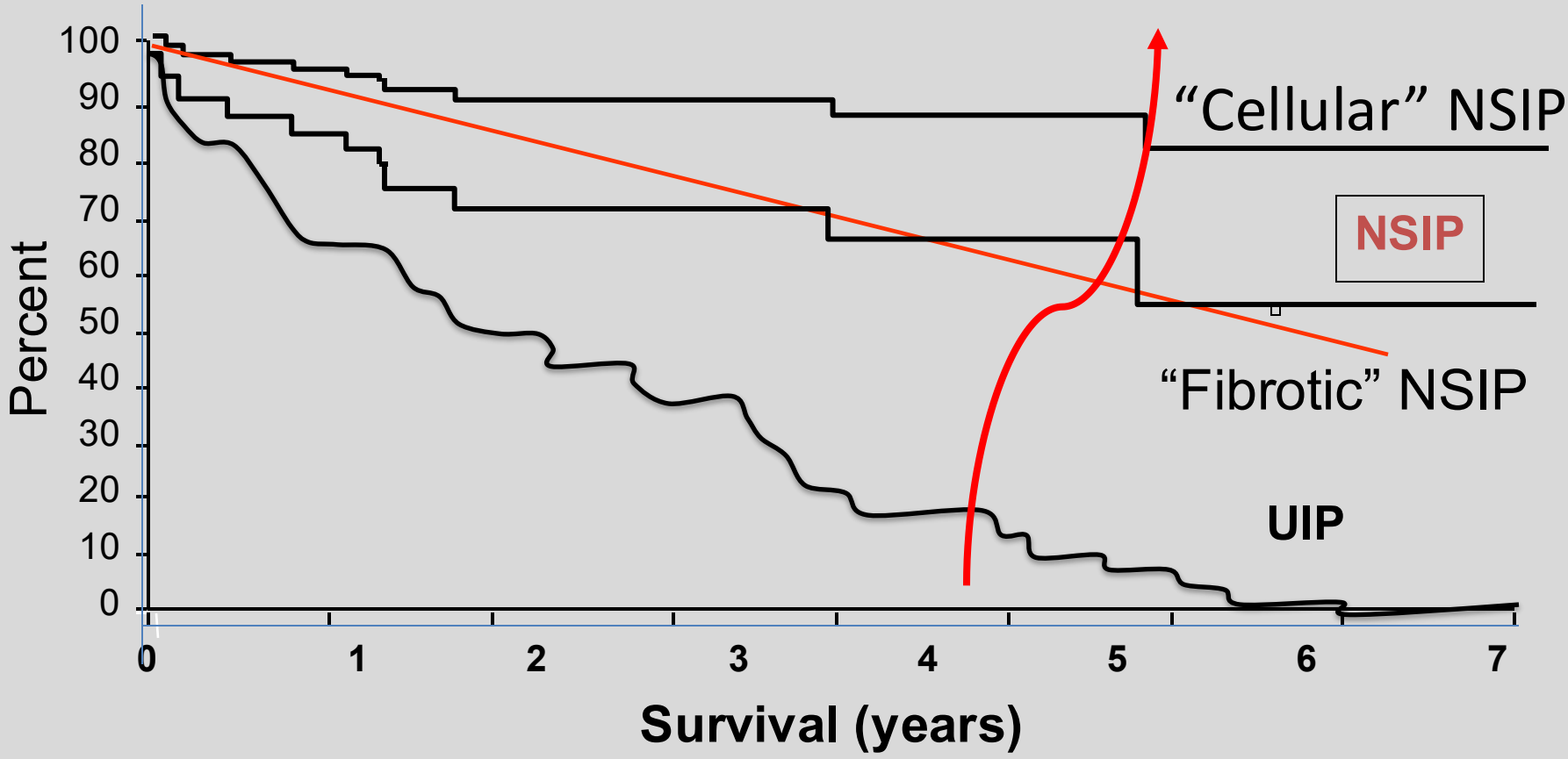
- Fine reticulation
- **Patchy ground glass** opacities



- **Temporal uniformity** on H&E stain
- w/ diffuse interstitial fibrosis and mild inflammatory infiltrate and no/few fibroblastic foci

ATS/ERS Consensus Statement. *Am J Respir Crit Care Med.* 2002;165:277-304
Schecter, M.D..

Survival of subtypes of NSIP vs UIP/IPF



Daniil ZD, et al. Am J Respir Crit Care Med. 1999;160:899-905.
Nagi S, et al. Eur Respir J. 1998;12:1010-1019.

Smoking Related Interstitial Disease

- Examples
  - **IPF** has strong correlation to smoking
  - **Pulmonary Langerhans Cell Histiocytosis** (other lecture) common in young smokers
  - We discuss two other relatively common forms:
    - Desquamative Interstitial Pneumonia **(DIP)**
    - Respiratory Bronchiolitis-Associated Interstitial Lung Disease **(RBILD)**

DIP

- Large collections of **pigmented macrophages** in airspace in a current/former smoker
  - Originally thought to be desquamated pneumocytes–hence the name
- Macrophages contain remnants of necrotic type 2 pneumocytes-**lamellar bodies** (contain surfactant)

- Usually presents 4-5th decade
- Sx/Sx: Dyspnea (**insidious** onset), **dry cough**, weeks-months, clubbing
  - PFTs – mild restrictive abnormality w/ moderate ↓ in diffusing capacity of CO (DLCO)

- Excellent response to **steroid Rx & cessation** of smoking, though some may go on to interstitial fibrosis

RBILD

- Chronic **inflammation** & peribronchiolar **fibrosis**
- Pigmented **intraluminal macrophages** (similar to DIP) in 1st & 2nd order **respiratory bronchioles**

- Usually **mild** symptoms, gradual onset of dyspnea and cough in current smokers, presentation usually 4th-5th decade
  - **Can be incidental** finding if mild, but called this if patient has significant lung symptoms, abnormal pulmonary function & imaging
- **Cessation of smoking** causes improvement

Thank you,

Lecture Feedback Form Link:

NYITCOM Student Feedback Form